

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Sequences

Read the trigger, name the data, choose the construction, write the bottom line.

SEQUENCES · CH₃ · L1

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The Map

This lesson is not a list of formulas. It is six exam moves. The question starts when you recognise the words that tell you what the working must become.

01 The rule (n th term)

TRIGGER *"nth term", "expression in terms of n ", "the rule"*

02 A named term or membership test

TRIGGER *"20th term", "is 150 a term?", "which term is ...?"*

03 The sum

TRIGGER *"sum", "total", "altogether", "from the 50th to the 100th"*

04 Two clues $\rightarrow a$ and d

TRIGGER *two facts about the same sequence, for example u_{20} and S_{30}*

05 Letters inside first terms

TRIGGER *terms like $8p$, $7p - 3$, $4p + 2$*

06 Context and geometry

TRIGGER *savings, shelves, polygons, counterexamples*

BANK EVIDENCE 30 unique solved source questions were mined from the Sequences bank, worth 134 marks. The largest weight is arithmetic series and algebraic sequence terms.

01 The Rule — n th Term

TRIGGER “ n th term”, “expression in terms of n ”, first few terms

BECOMES A rule that jumps straight from the position n to the term u_n .

FIRST LINE TO WRITE

$$d = \text{second term} - \text{first term} \quad \text{then} \quad u_n = a + (n - 1)d$$

SIMPLEST STRATEGY

- 1 Read the first term as a .
- 2 Find the common difference d .
- 3 Write $u_n = a + (n - 1)d$.
- 4 Expand and tidy.
- 5 Check by putting $n = 1$.

WORKED MODEL

First four terms: 85, 79, 73, 67.

$$d = 79 - 85 = -6$$

$$u_n = 85 + (n - 1)(-6) = 85 - 6n + 6 = 91 - 6n$$

BOTTOM LINE

$$u_n = 91 - 6n$$

SAME IDEA Find the n th-term rule from the first few terms.

QUESTION

Here are the first four terms of an arithmetic sequence:

38, 31, 24, 17

Find an expression, in terms of n , for the n th term of the sequence.

YOUR SOLUTION

02 A Named Term or a Membership Test

TRIGGER “the 20th term”, “which term is ...?”, “is 150 a term?”

BECOMES Either substitute a position into the rule, or solve the rule for n .

FIRST LINE TO WRITE

named term: substitute n

membership: solve $u_n = \text{given value}$

SIMPLEST STRATEGY

- 1 If the question says “20th”, put $n = 20$.
- 2 If it asks whether a value is in the sequence, set u_n equal to that value.
- 3 Solve for n .
- 4 The value is a term only if n is a positive whole number.

WORKED MODEL

The n th term is $4n + 7$. Show a counterexample to the rule $5n + 4$.

$$5(3) + 4 = 19$$

But 19 also appears in the first sequence:

$$4(3) + 7 = 19$$

BOTTOM LINE

A suitable counterexample is 19.

SAME IDEA Substitute a position, or solve the rule for n .

QUESTION

The n th term of a sequence is $5n + 17$.

(a) Find the 12th term.

(b) Is 152 a term in this sequence? Show your working.

YOUR SOLUTION

03 The Sum — Arithmetic Series

TRIGGER “sum”, “total”, “altogether”, S_n , “from term r to term s ”

BECOMES A series question. You are adding terms, not finding one term.

FIRST LINE TO WRITE

$$S_n = \frac{n}{2}(2a + (n-1)d) \quad \text{or} \quad S = \frac{\text{number}}{2}(\text{first} + \text{last})$$

SIMPLEST STRATEGY

- 1 Decide whether it is first- n or a block from one position to another.
- 2 For a block, find the first term of the block and the last term of the block.
- 3 Count the terms carefully: last position minus first position plus 1.
- 4 Use the first-plus-last version when both ends of the block are known.

WORKED MODEL

First terms: 8, 15, 22, 29, 36. Find the sum from the 50th term to the 100th term inclusive.

$$d = 7, \quad u_n = 8 + 7(n-1) = 7n + 1$$

$$u_{50} = 351, \quad u_{100} = 701, \quad \text{number} = 100 - 50 + 1 = 51$$

$$S = \frac{51}{2}(351 + 701) = 26826$$

— BOTTOM LINE

26826.

04 Two Clues $\rightarrow a$ and d

TRIGGER *two facts about the same arithmetic sequence, for example $u_{20} = 123$ and $S_{30} = 2880$*

BECOMES Two simultaneous equations in a and d .

FIRST LINE TO WRITE

$$u_r = a + (r - 1)d \quad \text{and} \quad S_n = \frac{n}{2}(2a + (n - 1)d)$$

SIMPLEST STRATEGY

- 1 Turn the term fact into equation 1.
- 2 Turn the sum fact into equation 2.
- 3 Simplify both equations.
- 4 Eliminate one letter.
- 5 Write both values before answering the final request.

WORKED MODEL

$$u_{20} = a + 19d = 123 \quad (1)$$

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880$$

$$2a + 29d = 192 \quad (2)$$

Double (1), then subtract (2):

$$2a + 38d = 246 \quad 9d = 54 \quad d = 6$$

$$a + 19(6) = 123 \quad a = 9$$

BOTTOM LINE

$$\boxed{a = 9, d = 6}.$$

SAME IDEA Turn two facts into two equations, then solve.

QUESTION

An arithmetic series has first term a and common difference d . The 21st term is 109. The sum of the first 52 terms is 4381. Work out the 5th term of the series.

YOUR SOLUTION

05 Letters Inside the First Terms

TRIGGER *terms such as $8p$, $7p - 3$, $4p + 2$*

BECOMES An equal-differences question first. The real sequence appears after the letter is found.

FIRST LINE TO WRITE

$$\text{term 2} - \text{term 1} = \text{term 3} - \text{term 2}$$

SIMPLEST STRATEGY

- 1 Use equal differences to solve the letter.
- 2 Replace the letter in the original terms.
- 3 Now identify a and d .
- 4 Continue with u_n or S_n , depending on what the question asks.

WORKED MODEL

The first three terms are $8p$, $7p - 3$, $4p + 2$.

$$(7p - 3) - 8p = (4p + 2) - (7p - 3)$$

$$-p - 3 = -3p + 5, \quad 2p = 8, \quad p = 4$$

So the sequence is:

$$32, \quad 25, \quad 18, \dots \quad a = 32, \quad d = -7$$

If $S_n = -1914$,

$$\frac{n}{2}(2(32) + (n-1)(-7)) = -1914$$

$$\frac{n}{2}(71 - 7n) = -1914 \quad \Rightarrow \quad 7n^2 - 71n - 3828 = 0$$

$$n = \frac{71 + 335}{14} = 29$$

BOTTOM LINE

$$\boxed{n = 29}.$$

SAME IDEA Use equal differences first, then continue with the real sequence.

QUESTION

The first three terms of an arithmetic sequence are

$$5q, \quad 4q + 3, \quad q + 12.$$

Find q . Then find the sum of the first 20 terms of the sequence.

YOUR SOLUTION

06 Context and Geometry

TRIGGER *savings by year, shelves, polygon angles, or one counterexample*

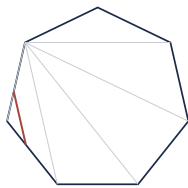
BECOMES A hidden sequence. Translate the story first, then use the same sequence rules.

FIRST LINE TO WRITE

name a, d, n then write the outside rule beside the sequence rule

SIMPLEST STRATEGY

- 1 Savings: count years first, especially when the years are inclusive.
- 2 Shelves or patterns: count each object type in terms of n .
- 3 Polygon angles: sequence sum equals $180(n - 2)$.
- 4 Counterexample: one valid value that breaks the claim is enough.



sequence of interior angles

$$\begin{array}{ccccccc}
 84^\circ & & 88^\circ & & 92^\circ & & \dots \\
 \bullet & \text{---} & \bullet & \text{---} & \bullet & \text{---} & \bullet \\
 & \text{---} & & \text{---} & & \text{---} &
 \end{array}$$

$$\begin{aligned}
 S_n &= \frac{n}{2}(2a + (n-1)d) \\
 S_n &= 180(n-2)
 \end{aligned}$$

FIGURE Match the angle-series sum with the polygon interior-angle sum.

WORKED MODEL

The interior angles of a polygon form an arithmetic sequence. The smallest angle is 84° , and the common difference is 4° .

$$S_n = \frac{n}{2}(2(84) + (n-1)4) = 2n^2 + 82n$$

The polygon interior sum is:

$$180(n-2)$$

Set equal:

$$2n^2 + 82n = 180(n-2)$$

$$n^2 - 49n + 180 = 0 \quad (n-45)(n-4) = 0$$

Since $n > 5$, $n = 45$, so

$$\text{sum} = 180(45-2) = 7740$$

BOTTOM LINE

7740°.

SAME IDEA Translate the story into a sequence rule and an outside rule.

QUESTION

The interior angles of a polygon form an arithmetic sequence. The smallest angle is 80° , and the common difference is 5° . The polygon has more than 5 sides. Work out the sum of the interior angles of the polygon.

YOUR SOLUTION

Quick Reference

TRIGGER → ACTION

IF YOU SEE	DO THIS FIRST
"nth term", "expression in terms of n "	$u_n = a + (n - 1)d$.
"the 20th term"	Substitute $n = 20$.
"is this value a term?"	Solve $u_n = \text{value}$ and check n is a positive integer.
"sum", "total", "altogether"	$S_n = \frac{n}{2}(2a + (n - 1)d)$.
"from term r to term s "	number of terms $= s - r + 1$.
two facts about the sequence	write two equations, then solve for a, d .
letters in first terms	equal differences first.
polygon angles	sequence sum $= 180(n - 2)$.

THE BOTTOM LINES

- 1 **Rule:** $u_n = 3n - 2$, or another clean expression in n .
- 2 **Term:** one value, such as $u_{20} = 77$.
- 3 **Membership:** the position n , or not a term.
- 4 **Sum:** one total, with units if the problem has units.
- 5 **Two clues:** a and d , then the exact thing asked for.
- 6 **Context:** the story answer, not only the algebra.

FORMULA BANK

$$u_n = a + (n - 1)d \quad d = u_2 - u_1 \quad S_n = \frac{n}{2}(2a + (n - 1)d) \quad S = \frac{\text{number}}{2}(\text{first} + \text{last})$$

$$180(n - 2)$$

PRACTICE ORDER

Linear nth term → block sums → two-equation questions → letters/context/geometry.